

Assignment 3

Combinational- Assignment

1. Implement the following functions using i) 2:1 MUX and NOT gates ii) 4:1 MUX

$$\text{a) } F(A, B, C) = \bar{A}C + B\bar{C}$$

$$\text{b) } F(P, Q, R) = \bar{P}\bar{R} + Q\bar{R} + \bar{P}Q$$

2. Implement the functionality of a full adder using 2:1 Muxes only.
3. Simplify $f(a, b, c, d) = \sum m(2, 4, 5, 6, 10) + D(12, 13, 14, 15)$ and implement the SOP expression using 4:1 MUX.